

Generative Modeling

Advanced Normalizing flows

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Laboratory of methods for big data analysis



Spring 2025

In this Lecture

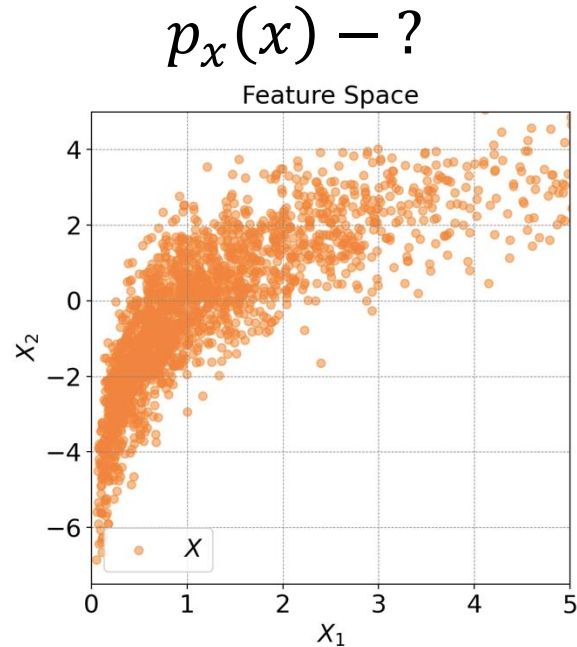
- ▶ Ordering in block models.
- ▶ Full Jacobian models.
- ▶ Continuous time models.

Reminder: Normalizing Flows



Problem Statement

$x_i \sim p_x(x)$ – data



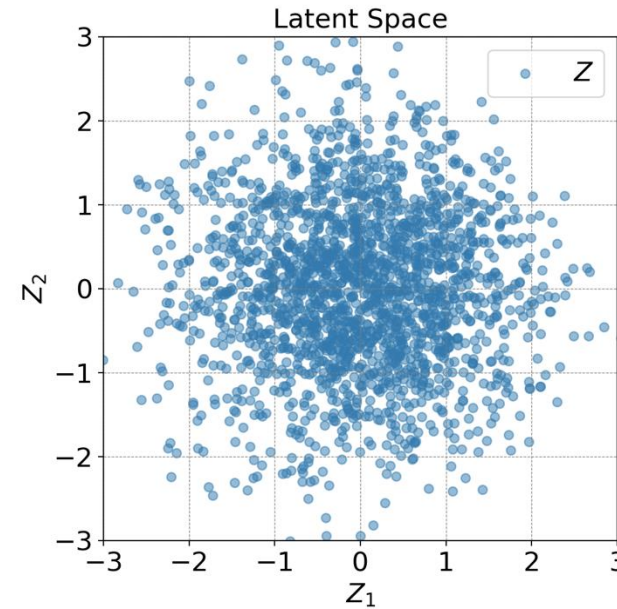
$$z = f(x) - ?$$



$$x = f^{-1}(z) - ?$$

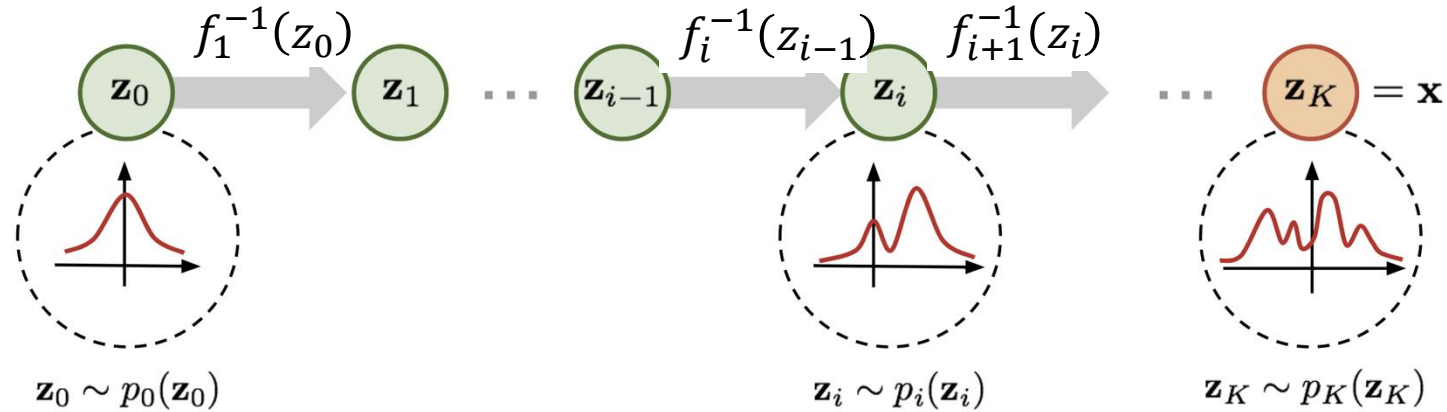
$z_i \sim p_z(z)$

$p_z(z)$ known



- **We have:** real objects $\{x_i\}$
- **Task:** find invertible and differentiable $f: z_i = f(x_i)$, such that $z_i \sim p_z(z)$. For some known: $p_z(z)$.

Flow discussion



$$p_x(x) = p_{z_0}(f_K(\dots(f_1(x)))) \left| \det \frac{\partial f_1(z_1)}{\partial z_1} \right| \dots \left| \det \frac{\partial f_K(z_k)}{\partial z_k} \right|.$$

- ▶ It is possible to obtain $p_x(x)$ consecutively changing observables.
- ▶ The overall transformation is invertible if individual layers are invertible.
- ▶ Dimensions of each observable is the same.
- ▶ Fit is performed using ML estimate.
- ▶ Need to calculate determinant.

Block matrix for Jacobian

$$\frac{\partial \mathbf{f}(x)}{\partial x} = \begin{pmatrix} \mathbb{I}_d & 0 \\ \frac{\partial z_{d+1:D}}{\partial x_{d+1:D}} & S \end{pmatrix}$$

- ▶ Randomly choose d such that we have two disjoint subsets of observables: $z_{1:d}$ and $z_{d+1:D}$.
- ▶ Insert a block transform.
- ▶ Repeat for several layers.
- ▶ If needed insert scaling layers.
- ▶ Fit simultaneously.

Real-NVP

$$z = \textcolor{red}{f}(x) = \begin{cases} z_{1:d} = x_{1:d} \\ z_{d+1:D} = x_{d+1:D} \odot \exp(\textcolor{blue}{s}(x_{1:d})) + \textcolor{blue}{t}(x_{1:d}) \end{cases}$$

- ▶ $\textcolor{blue}{s}(x_{1:d})$ и $\textcolor{blue}{t}(x_{1:d})$ – **neural networks** with d inputs and $D - d$ outputs.
- ▶ Invertible.
- ▶ $\det J_k = \exp \sum_{i=d+1}^D (\alpha_\theta(z_{1:d}))_i$ for k-th layer.
- ▶ Inspired by RNADE.

<https://arxiv.org/abs/1605.08803>

Jacobian Choices

1. Det Identities

Planar NF
Sylvester NF
...

Jacobian



(Low rank)

2. Coupling Blocks

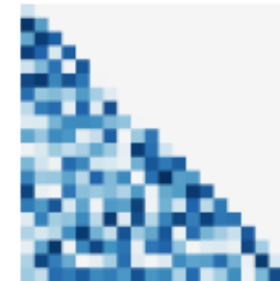
NICE
Real NVP
Glow
...



(Lower triangular +
structured)

3. Autoregressive

Inverse AF
Neural AF
Masked AF
...



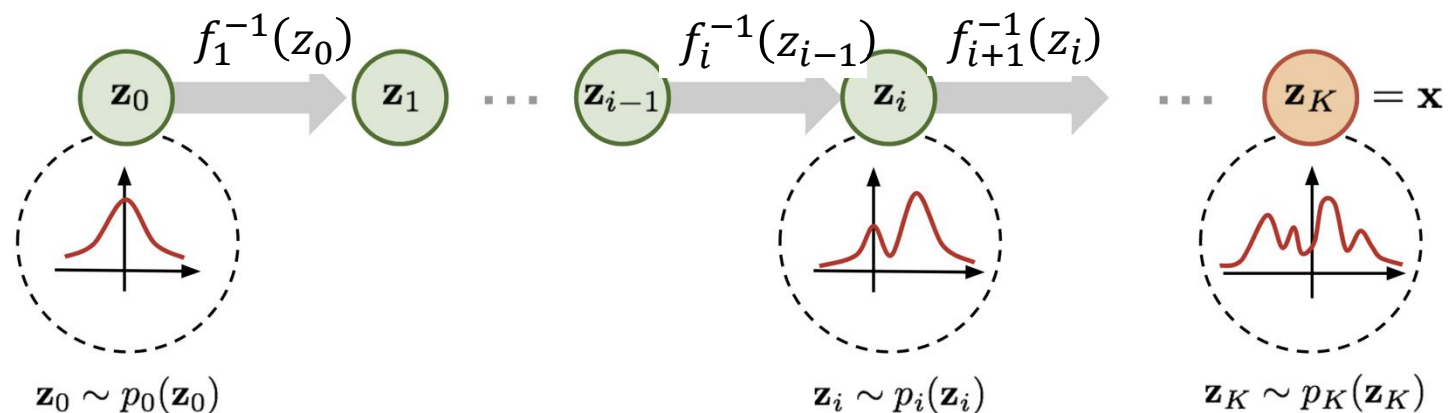
(Lower triangular)

Generative Flow with Invertible 1×1 Convolutions

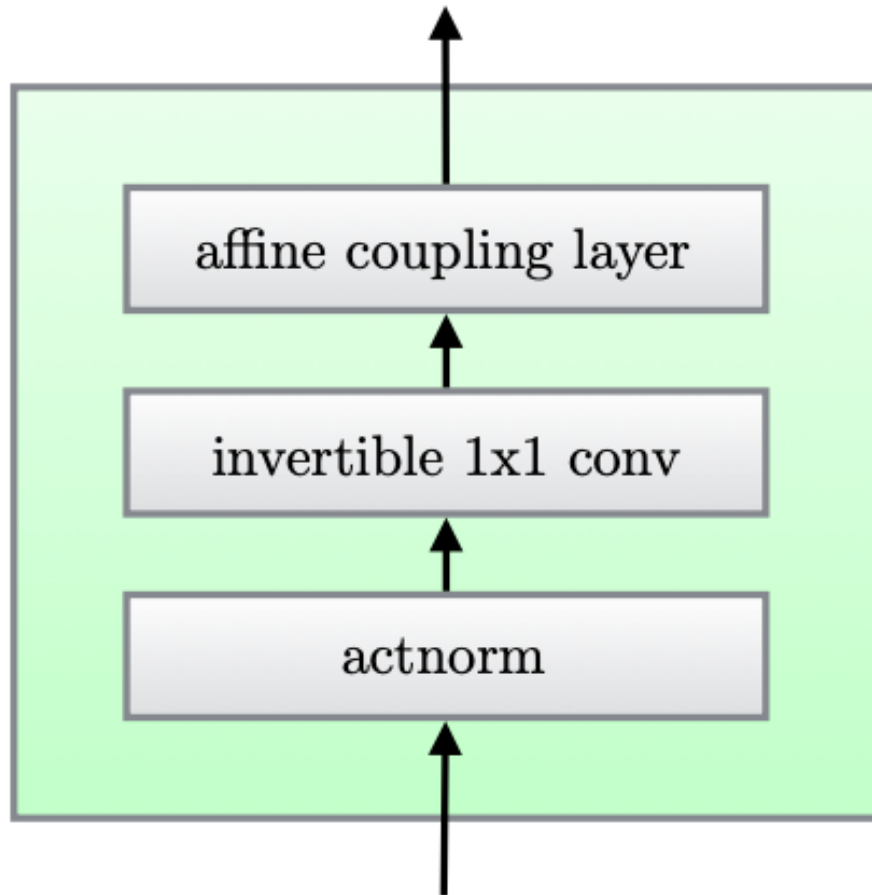


Motivation

- ▶ rNVP needs permutations to include all dimensions into consideration.
- ▶ This requires additional layers and several permutations.



One step of flow



► Activation normalization.

- Trainable scale and bias

$$z_{i,j} = \mathbf{s} \odot x_{i,j} + \mathbf{b}.$$

► Invertible 1x1 convolution.

- Trainable weight matrix.

$$z_{i,j} = \mathbf{W} x_{i,j}.$$

► Affine coupling layer.

- Similar to rNVP.

$$x_a, x_b = \text{split}(x)$$

$$z_a = \mathbf{s} \odot x_a + \mathbf{t}.$$

$$z_b = x_b.$$

<https://arxiv.org/pdf/1807.03039.pdf>

1x1 convolution layer

- ▶ Permutation is just a special case of a linear operator:

$$y = Wx = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} x.$$

- ▶ Convolution of an input $h \times w \times c$ tensor $\mathbf{h}$ with a weight matrix W ($c \times c$):

$$f = \text{conv2d}(h; W).$$

- Need change of variables formula:

$$\log \left| \det \left(\frac{df}{dh} \right) \right| = \log \left| \det \left(\frac{d\text{conv2D}(\mathbf{h}; W)}{dh} \right) \right| = h \cdot w \log \det |W|$$

<https://arxiv.org/pdf/1807.03039.pdf>

Jacobian calculations

- ▶ PLU decomposition:

$$W = PL(U + \text{diag}(s)).$$

$$\log |\det W| = \sum \log |s|.$$

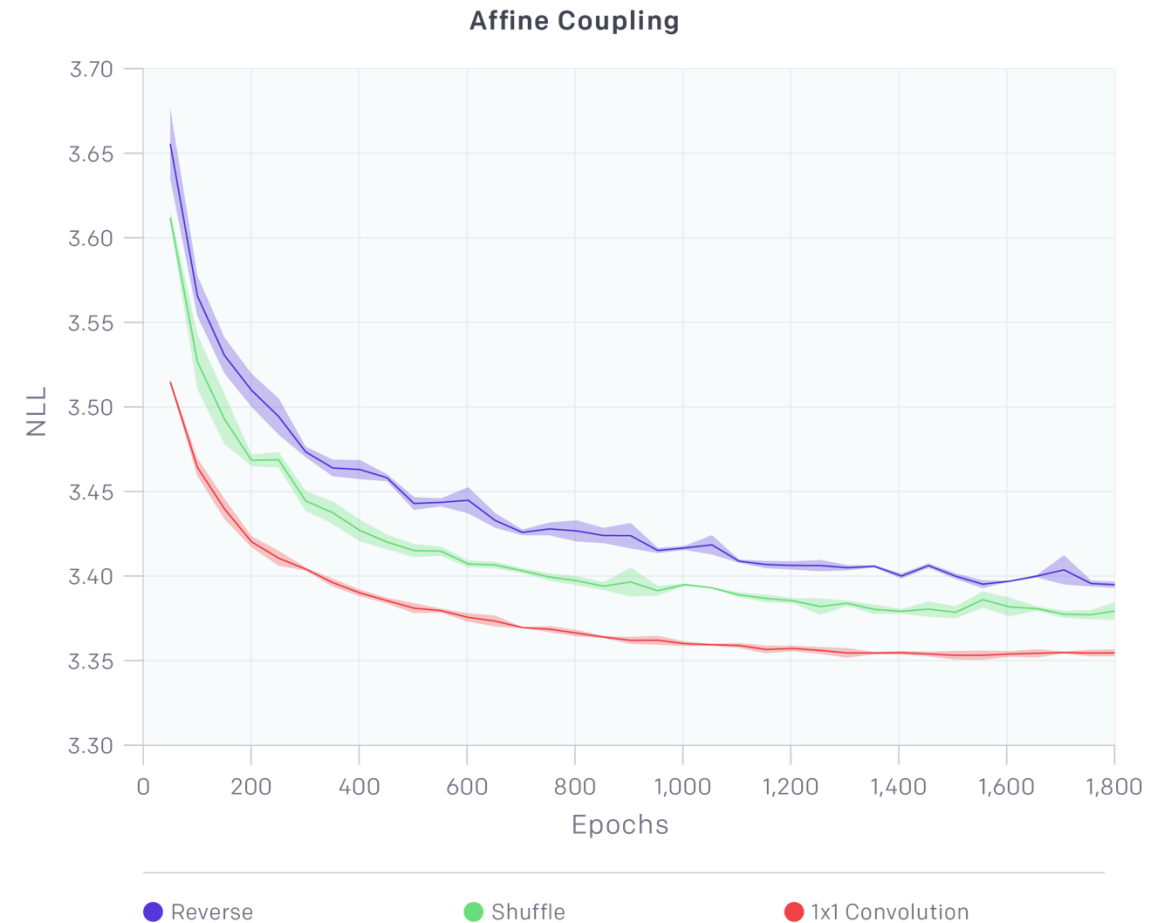
Initiate from random W .

P is permutation (remains fixed), L , U , and s are optimized

<https://arxiv.org/pdf/1807.03039.pdf>

Affine coupling

- ▶ Faster to converge than additive.
- ▶ 1x1 convolution performs like better randomization.



Sampling Temperature

- ▶ In order to get more realistic sampling, one can use a reduced-temperature model.
- ▶ In this work:

$$p_{\theta,T}(x) \sim p_{\theta}^{T^2}(x)$$

Temperature is a free parameter for sampling.

<https://arxiv.org/abs/1702.00783>

Temperature Dependence



Figure 8: Effect of change of temperature. From left to right, samples obtained at temperatures 0, 0.25, 0.6, 0.7, 0.8, 0.9, 1.0

GLOW: results

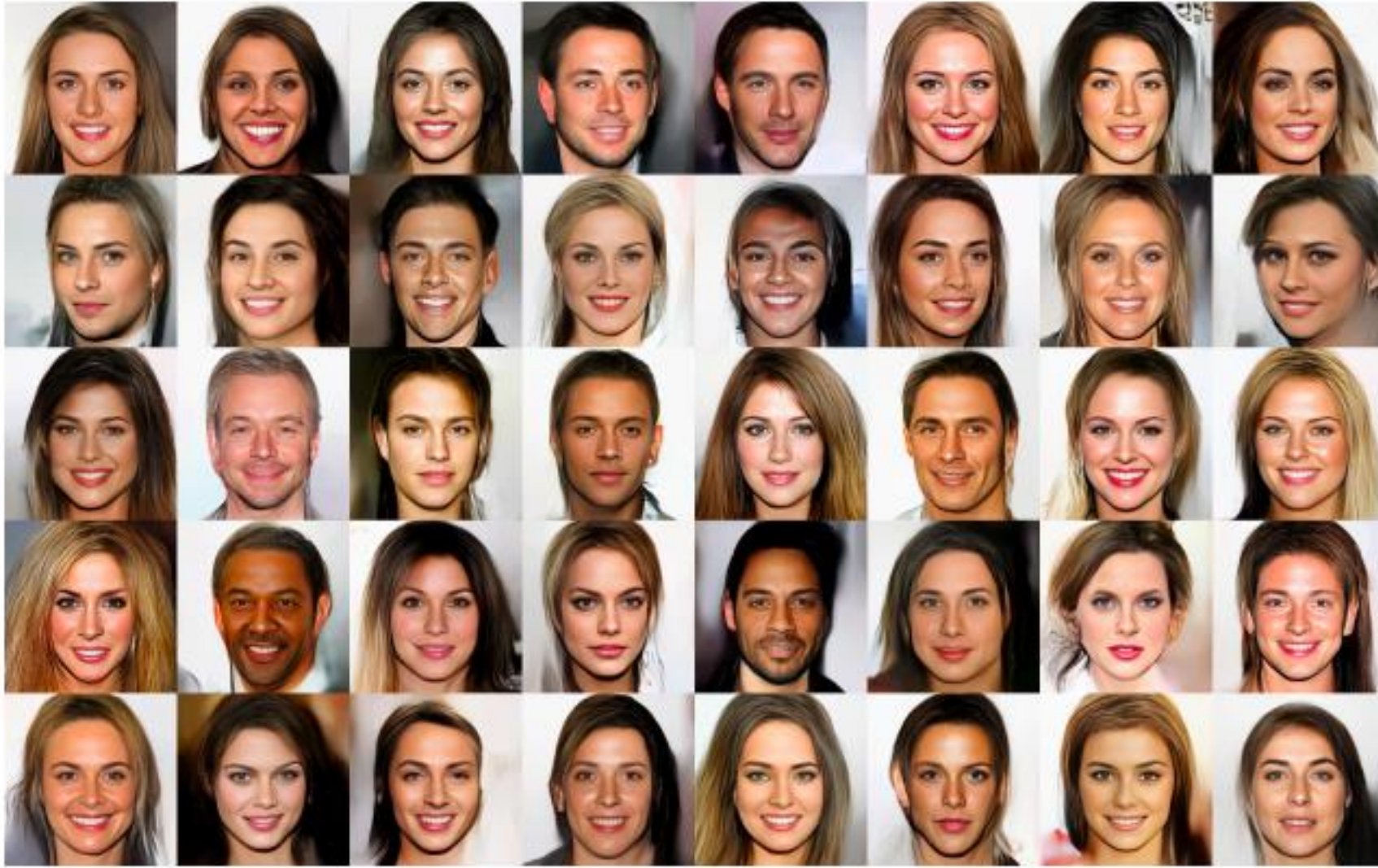


Figure 4: Random samples from the model, with temperature 0.7

GLOW: depth dependence

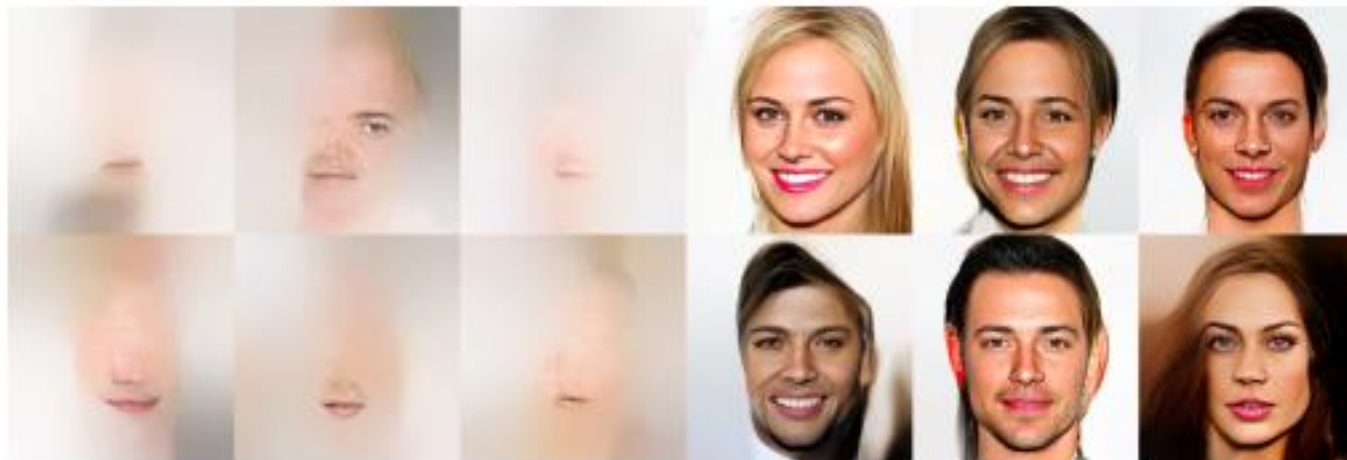


Figure 9: Samples from shallow model on left vs deep model on right. Shallow model has $L = 4$ levels, while deep model has $L = 6$ levels

Conclusions

- ▶ Addresses problem of choosing permutation with 1x1 convolutional layer.
- ▶ Uses triangular Jacobian idea.
- ▶ Quite slow.

<https://arxiv.org/abs/1702.00783>

More Linear Flows

- ▶ QR Flows:

$$W = QR$$

Q is orthogonal, R is upper triangular.

$$\det W = \prod R_{ii}$$

- ▶ Orthogonal flows

W – orthogonal, $\det W = 1$

<https://arxiv.org/pdf/1912.02762.pdf>

Residual Flows



Jacobian Choices

1. Det Identities

Planar NF
Sylvester NF
...

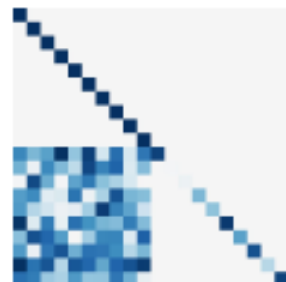
Jacobian



(Low rank)

2. Coupling Blocks

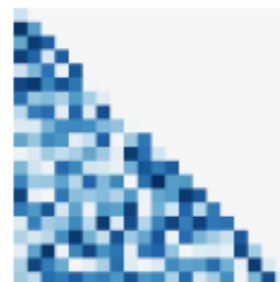
NICE
Real NVP
Glow
...



(Lower triangular +
structured)

3. Autoregressive

Inverse AF
Neural AF
Masked AF
...



(Lower triangular)

4. Unbiased Estimation

FFJORD
Residual Flows



(Arbitrary)

Invertible Residual Networks (i-ResNet)

Residual blocks:

$$y = F(x) = x + \textcolor{brown}{g}(x)$$

can be inverted by **fixed point iteration**:

$$x = y - \textcolor{brown}{g}(x)$$

and has unique inverse if

$$\text{Lip}(\textcolor{brown}{g}) < 1$$

Thus, effectively making $\textcolor{brown}{g}$ contractive.

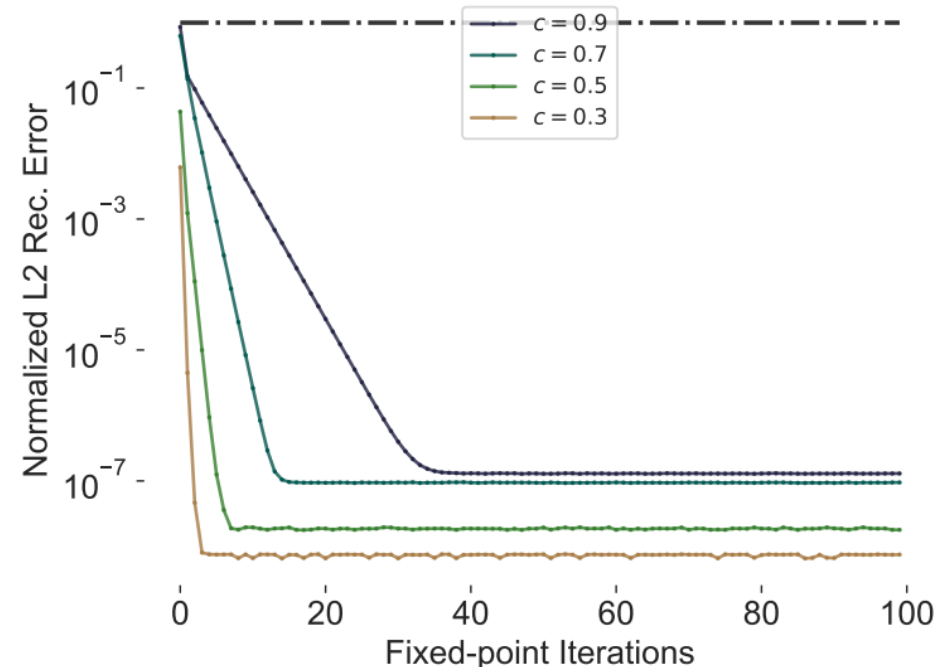
Algorithm 1. Inverse of i-ResNet layer via fixed-point iteration.

Input: output from residual layer y , contractive residual block g , number of fixed-point iterations n

Init: $x^0 := y$

for $i = 0, \dots, n$ **do**
 $x^{i+1} := y - g(x^i)$

end for



Invertible Residual Networks (i-ResNet)

Residual blocks:

$$y = F(x) = x + \textcolor{red}{g}(x)$$

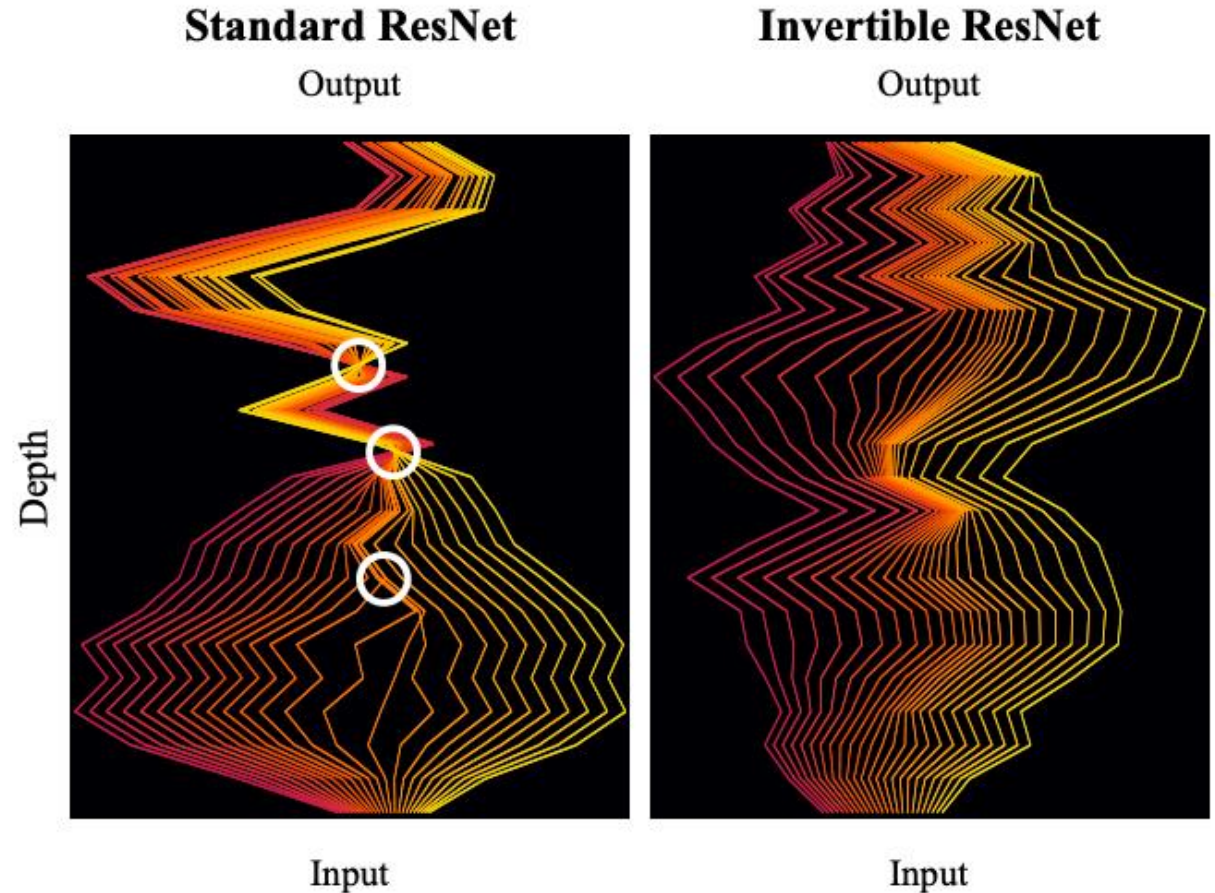
can be inverted by **fixed point iteration**:

$$x = y - \textcolor{red}{g}(x)$$

and has unique inverse if

$$\text{Lip}(\textcolor{red}{g}) < 1$$

Thus, effectively making $\textcolor{red}{g}$ contractive.



Satisfying Lipschitz Condition on $g(x)$

- ▶ Construct g as a network

$$g = W_3 \circ \phi \circ W_2 \circ \phi \circ W_1 \circ \phi.$$

- ▶ Can construct data independent Lip-constraint:

$$\text{Lip}(g) < \|W_3\|_2 \|W_2\|_2 \|W_1\|_2$$

- ▶ Use spectral norm of weight matrices (σ the largest singular layer):

$$\tilde{W} = \frac{cW}{\sigma}$$

<https://arxiv.org/abs/1811.00995>

Satisfying Lipschitz Condition on $g(x)$



Figure 3. Original images (top) and reconstructions from i-ResNet with $c = 0.9$ (middle) and a standard ResNet with the same architecture (bottom), showing that the fixed point iteration does not recover the input without the Lipschitz constraint.

<https://arxiv.org/abs/1811.00995>

Change of Variables Formula

$$\log p_X(x) = \log \mathbf{p}_Z(F(x)) + \log |\det J_F(x)|$$

- ▶ Need a way to compute Jacobian:

$$F = (I + g)x.$$

- ▶ Insert into the calculations and $\ln \det A = \text{tr}(\log A)$:

$$\log p_X(x) = \log \mathbf{p}_Z(F(x)) + \text{tr}(\log(I + J_g(x))).$$

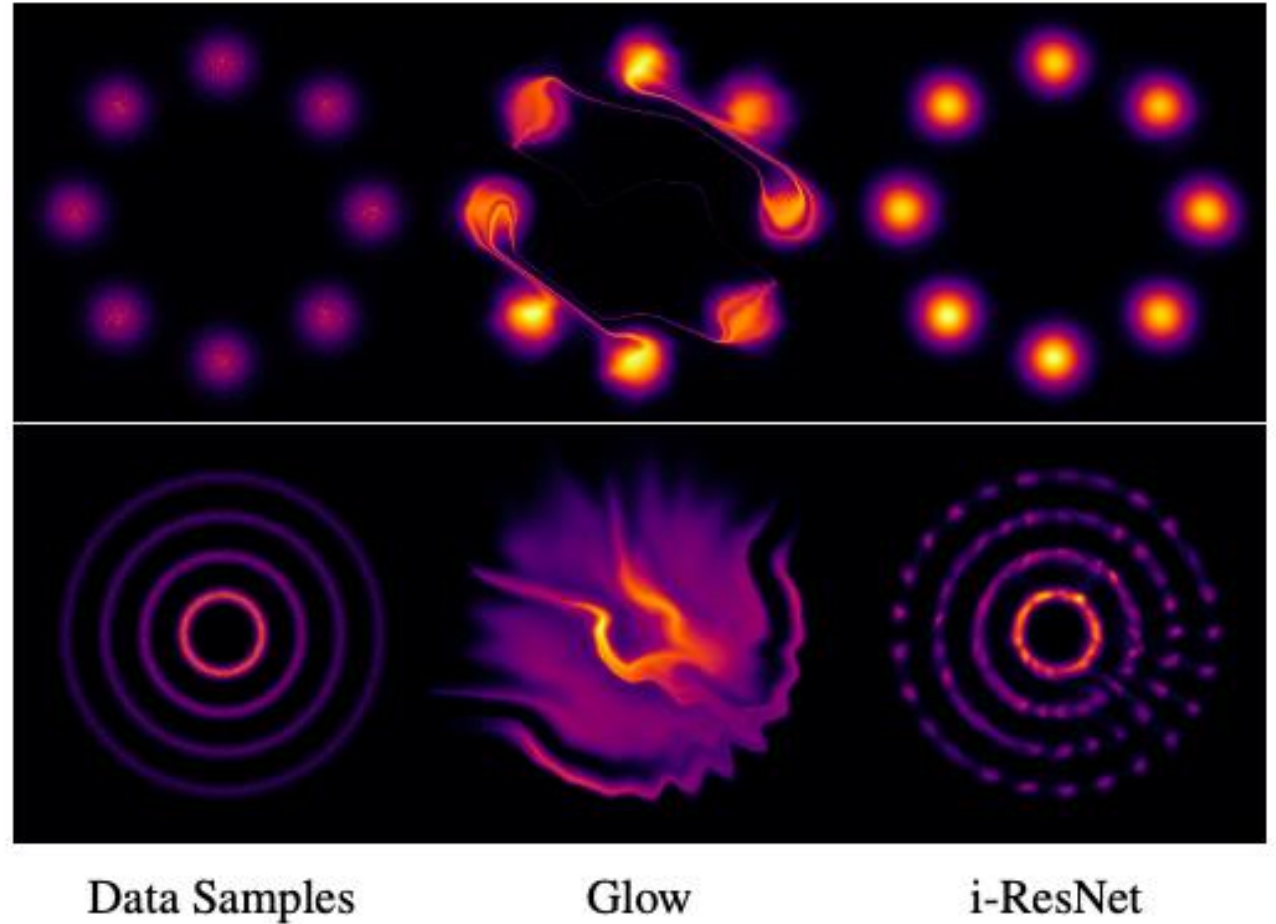
- ▶ Trace of the matrix logarithm can be expressed as power series (for constrained $\|J_g\| < 2$):

$$\log p_X(x) = \log \mathbf{p}_Z(F(x)) + \sum_{k=1}^{\infty} (-1)^{k+1} \frac{\text{tr}(J_g^k)}{k}.$$

- ▶ The sum can be truncated and reduced to biased Hutchinson estimate.

Invertible Residual Networks (i-ResNet)

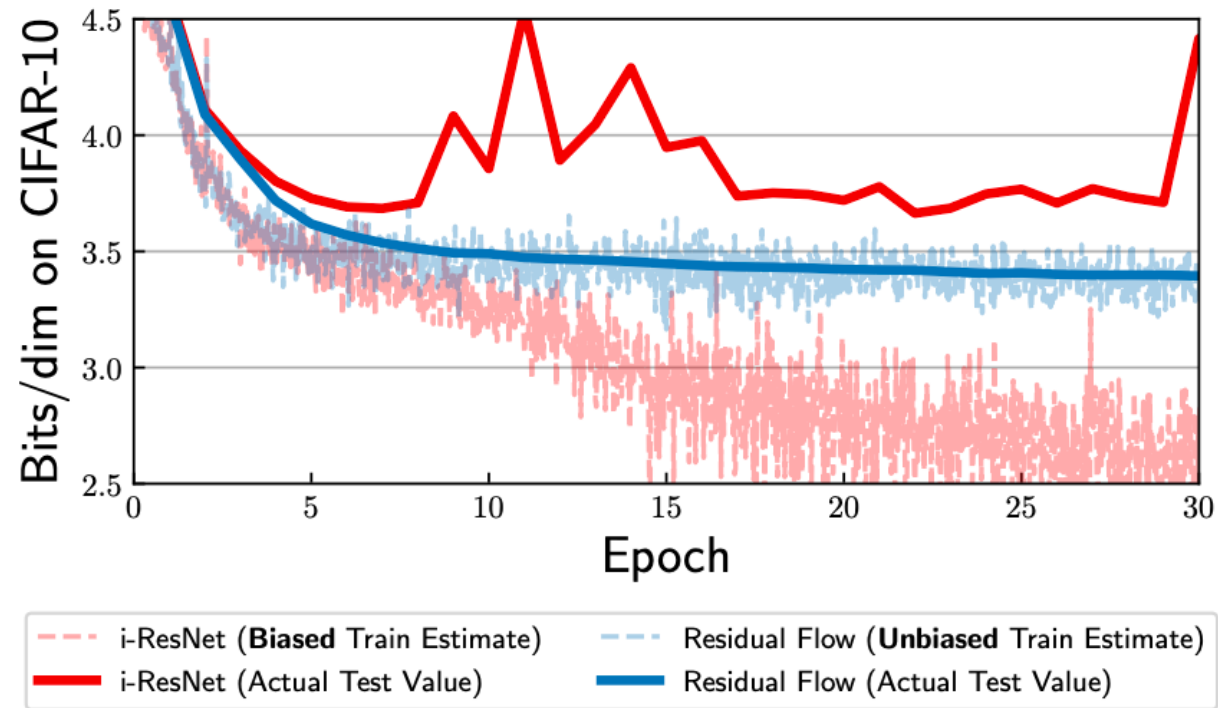
- ▶ Performs better than glow.
- ▶ Still some troubles are visible.
- ▶ Can we estimate in a different way?



Residual Flows

- Use Russian roulette estimate for infinite series:

$$\sum_{k=1}^{\infty} \Delta_k = \mathbb{E}_{n \sim p(N)} \left[\sum_{k=1}^n \frac{\Delta_k}{\mathbb{P}(N \geq k)} \right]$$



<https://arxiv.org/pdf/1906.02735.pdf>

Residual Flows results



Figure 5: **Qualitative samples.** Real (left) and random samples (right) from a model trained on 5bit 64×64 CelebA. The most visually appealing samples were picked out of 5 random batches.

Table 2: Lower FID implies better sample quality. *Results taken from [Ostrovski et al. \(2018\)](#).

Model	CIFAR10 FID
PixelCNN*	65.93
PixelIQN*	49.46

i-ResNet	65.01
Glow	46.90
Residual Flow	46.37

DCGAN*	37.11
WGAN-GP*	36.40

- ▶ Performs better than many flow models.
- ▶ Still worse than GANs

Residual Flows Discussion

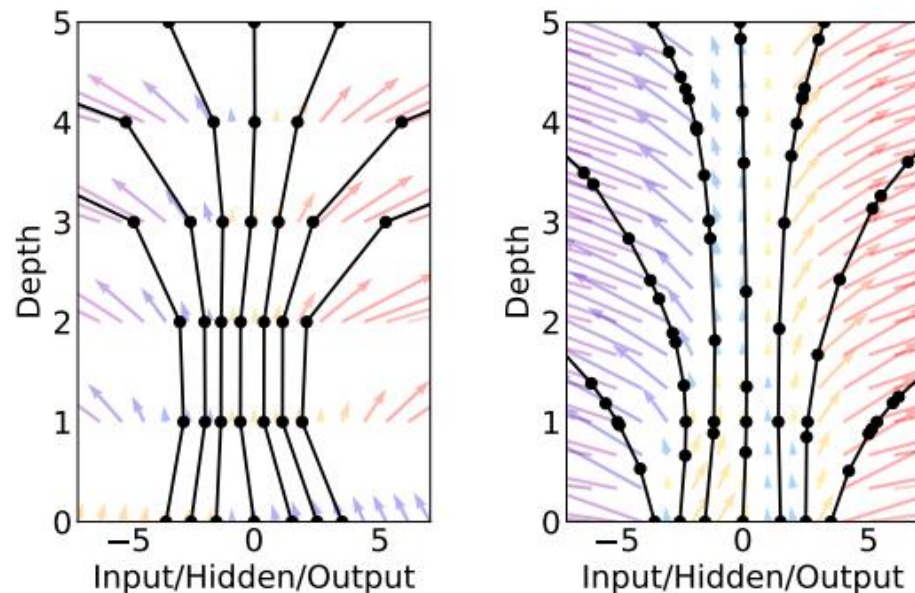
- ▶ RF has dense Jacobian, thus are more flexible than AR flows.
- ▶ Density evaluation needs to be done iteratively, thus is slow.

https://www.cs.toronto.edu/~rtqichen/pdfs/residual_flows_slides.pdf

FFJORD



Motivation

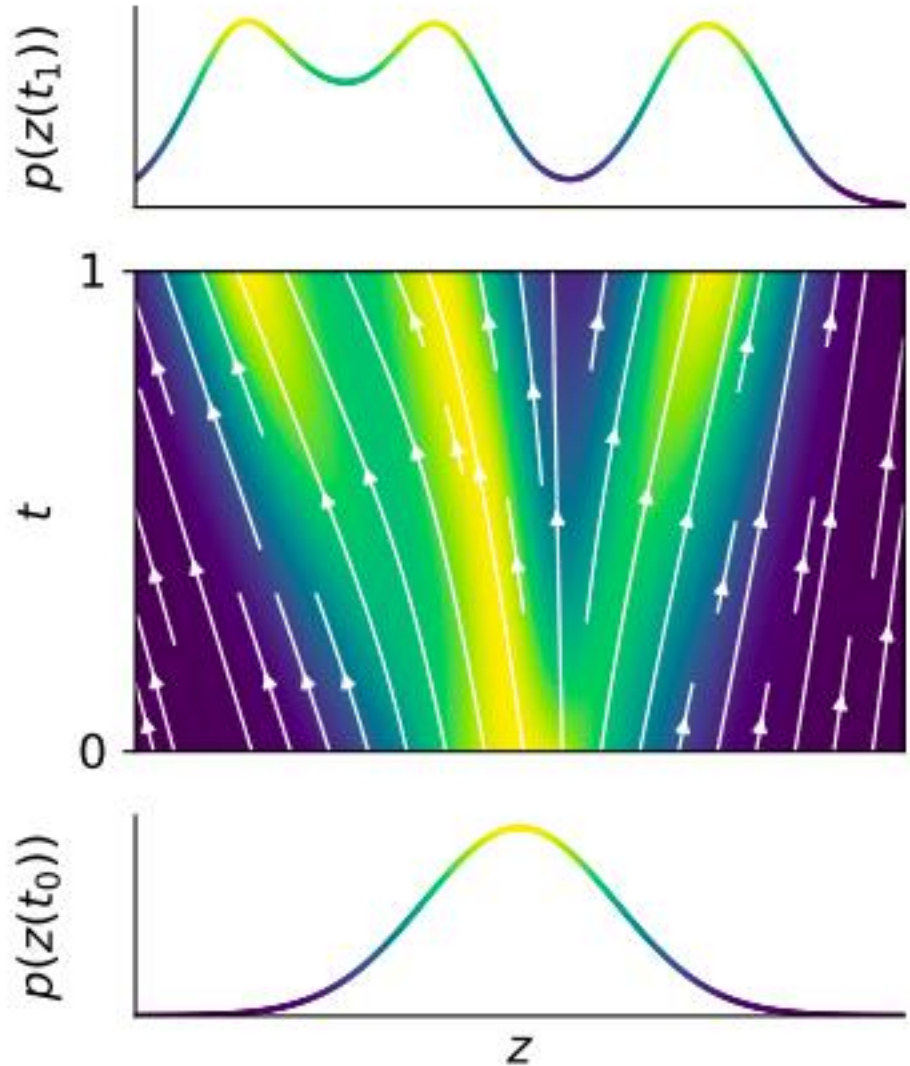


- ▶ Do we really care of having discrete steps?
- ▶ Can we change the Jacobian to something more stochastic?
- ▶ System of continuous-time dynamics.
- ▶ These ideas led to the NeuralODE model.

<https://arxiv.org/abs/1806.07366>

<https://stanniszhou.github.io/discussion-group/post/ffjord/>

Continuous Normalizing Flows



- Model generative process with continuous dynamics:

$$z_0 \sim p(z_0)$$

$$\frac{\partial z}{\partial t} = f_{\theta}(z_t, t)$$

$$x = z_t = z_0 + \int_{t_0}^{t_1} f_{\theta}(z_t, t) dt$$

Instantaneous Change-of-variable Formula

- ▶ For f uniformly Lipschitz continuous in $\mathbf{z}$ and continuous in t

$$\frac{\partial \log p(\mathbf{z}(t))}{\partial t} = -\text{tr} \left(\frac{df}{d\mathbf{z}}(t) \right)$$

- ▶ The complete initial value problem is then given as

$$\frac{d}{dt} \begin{bmatrix} \mathbf{z}(t) \\ \log p(\mathbf{z}(t), t) \end{bmatrix} = \begin{bmatrix} f(\mathbf{z}(t), t, \boldsymbol{\theta}) \\ -\text{tr} \partial_{\mathbf{z}} f \end{bmatrix}, \quad t \in [0, T],$$

$$\mathbf{z}(0) \sim p_{\mathbf{z}_0}(\mathbf{z}(0)),$$

$$\log p(\mathbf{z}(0), 0) = \log p_{\mathbf{z}_0}(\mathbf{z}(0)).$$

Unbiased Log-Density Estimation

- ▶ log-probability of the data under continuous model

$$\log(p_x) = \log(p_{z_0}) - \int_0^1 \text{Tr} \frac{\partial f(z(t))}{\partial z(t)} dt$$

- ▶ This gives $O(N^3)$ calculations.
- ▶ Need a smart trick to increase the speed of calculations.

$$\begin{aligned} \log p(\mathbf{z}(t_1)) &= \log p(\mathbf{z}(t_0)) - \int_{t_0}^{t_1} \text{Tr} \left(\frac{\partial f}{\partial \mathbf{z}(t)} \right) dt \\ &= \log p(\mathbf{z}(t_0)) - \int_{t_0}^{t_1} \mathbb{E}_{p(\epsilon)} \left[\epsilon^T \frac{\partial f}{\partial \mathbf{z}(t)} \epsilon \right] dt \\ &= \log p(\mathbf{z}(t_0)) - \mathbb{E}_{p(\epsilon)} \left[\int_{t_0}^{t_1} \epsilon^T \frac{\partial f}{\partial \mathbf{z}(t)} \epsilon dt \right] \end{aligned}$$

FFJORD: results

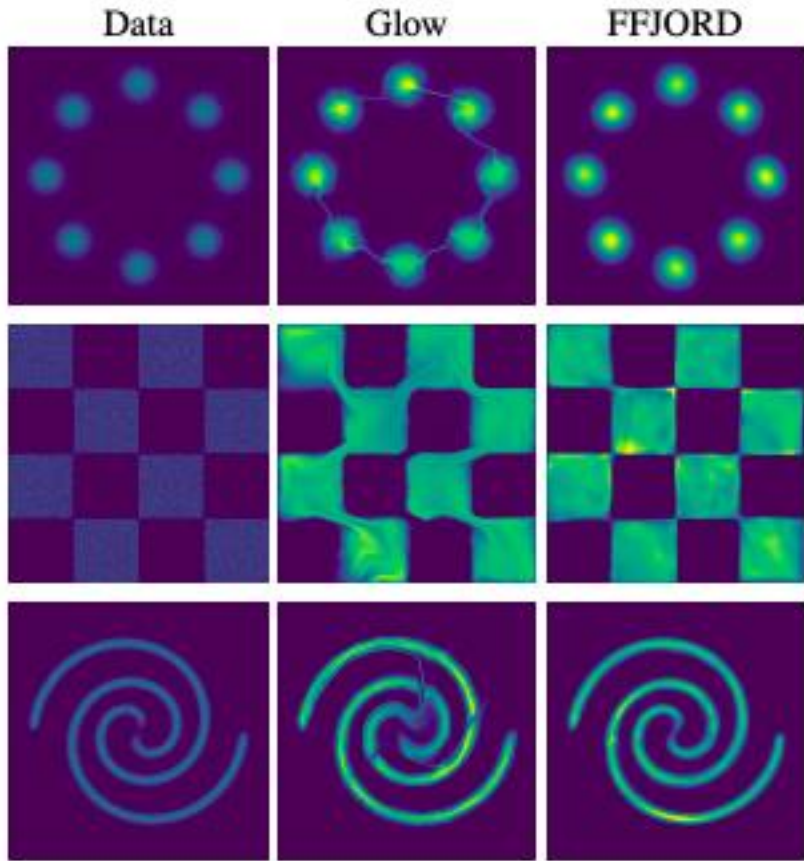


Figure 2: Comparison of trained FFJORD and Glow models on 2-dimensional distributions including multi-modal and discontinuous densities.

	POWER	GAS	HEPMASS	MINIB	BSDS	MNIST	CIFAR10
Real NVP	-.17	-8.33	18.71	13.55	-153.28	1.06	3.49
Glow	-.17	-8.15	18.92	11.35	-155.07	1.05	3.35
FFJORD	-.46	-8.59	15.26	10.43	-157.67	0.99*	3.40
MADE	3.08	-3.56	20.98	15.59	-148.85	2.04	5.67
MAF	-.24	-10.08	17.70	11.75	-155.69	1.89	4.31
TAN	-.48	-11.19	15.12	11.01	-157.03	-	-
DDSF	-.62	-11.96	15.09	8.86	-157.73	-	-

Table: Density estimation experiments. Negative log-likelihood on test set.

- Performs better than many flow models.

FFJORD: discussion

► Advantages

- Guaranteed inverse regardless of model parameterization
- Efficient, unbiased log-probability estimation without restricting the Jacobian of the transformation
- Reversible generative models can now be defined with standard neural network architectures.

► Disadvantages

- Relies on adaptive numerical ODE solvers for stable training
- Computation time determined by solver, not user
- 4-5x slower than other reversible generative models (Glow, Real-NVP)

FFJORD: discussion

Method		Train on data	One-pass Sampling	Exact log-likelihood	Free-form Jacobian
Change of Variables	Variational Autoencoders	✓	✓	✗	✓
	Generative Adversarial Nets	✓	✓	✗	✓
	Likelihood-based Autoregressive	✓	✗	✓	✗
	Normalizing Flows	✗	✓	✓	✗
	Reverse-NF, MAF, TAN	✓	✗	✓	✗
	NICE, Real NVP, Glow, Planar CNF	✓	✓	✓	✗
	FFJORD	✓	✓	✓	✓

Table 1: A comparison of the abilities of generative modeling approaches.

Conclusions



Normalizing flows: conclusions

- ▶ Advantages:
 - Explicit likelihood.
 - Straightforward sampling.
- ▶ Disadvantages:
 - Less realistic samples than GANs.
 - Some models have suppressed one of the advantages.